Categorical Sampling as an Intermediate Activation Function: Softmax Sampling Activation

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Abstract

Most modern neural networks rely on deterministic activation functions like ReLU, which lack inherent stochasticity for regularization or uncertainty estimation. While stochastic methods like Dropout or noise-based activations exist, the potential of using categorical sampling at intermediate layers remains largely unexplored. In this work, we propose Softmax Sampling Activation (SSA), a technique that treats intermediate activations as logits and samples a neuron to pass forward according to its relative probability. We leverage the Gumbel-Softmax reparameterization trick to enable differentiable categorical sampling within the network architecture. Our experiments on CIFAR-10 using a VGG-11 architecture reveal that while applying SSA at every layer leads to vanishing signals and training instability, selective application at the final feature layer allows for effective training. We show that "soft" SSA (continuous relaxation) achieves a test accuracy of 83.09%, compared to 86.46% for a standard ReLU baseline, whereas "hard" SSA (discrete sampling) fails to converge. Our analysis suggests that the competitive nature of categorical sampling creates a bottleneck that, while regularizing, can restrict representational capacity. We conclude that SSA is a viable but challenging primitive that provides a natural mechanism for stochastic feature selection.

1 Introduction

Why do we use deterministic activations? Most deep learning architectures rely on fixed, deterministic non-linearities such as ReLU or Swish to build hierarchical representations. While these functions are computationally efficient and enable gradient flow, they do not inherently capture the uncertainty or competition that might be beneficial during feature extraction. In contrast, the final layer of a classification network often uses a softmax distribution to represent a probability distribution over classes, effectively forcing a competition between mutually exclusive outcomes.

The introduction of stochasticity into neural networks has traditionally focused on regularization (e.g., Dropout Srivastava et al. [2014]) or modeling weight uncertainty (e.g., Bayesian Neural Networks). More recently, noise-based activations like PROBACT Shridhar et al. [2019] have shown that sampling from a continuous distribution at the neuron level can improve generalization. However, there is a significant gap in exploring **categorical sampling** as an intermediate primitive. Specifically, what happens if we treat an intermediate layer's activations as logits and sample a subset of "active" neurons according to their relative importance?

In this paper, we propose SSA (SSA), a novel activation module that introduces categorical competition at intermediate layers. SSA treats the pre-activations as logits, applies a softmax to generate a probability distribution, and samples according to this distribution to produce the layer output. To ensure differentiability, we employ the Gumbel-Softmax reparameterization trick Jang et al. [2017], allowing the network to learn through the sampling process.

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Our results on CIFAR-10 with a VGG-11 architecture demonstrate both the promise and the pitfalls of this approach. While naive application of SSA at every layer causes the signal to vanish exponentially, leading to 10.21% accuracy, a hybrid approach applying SSA only at the final feature layer achieves a respectable 83.09% accuracy. This is slightly below the 86.46% baseline set by ReLU, highlighting a fundamental trade-off between the flexibility of independent activations and the bottleneck of categorical competition.

Our main contributions are as follows:

- We propose SSA (SSA), a differentiable categorical sampling module for intermediate neural network layers.
- We identify the "vanishing signal" problem inherent in deep SSA stacks and provide an empirical characterization of its impact on convergence.
- We conduct a comparative study between "soft" relaxation and "hard" discrete sampling, demonstrating that continuous relaxation is essential for stabilizing the training of competitive activations.
- We analyze the information bottleneck created by SSA, suggesting that while it acts as a strong regularizer, it can also restrict the model's representational capacity compared to standard monotonic activations.

2 Related Work

Stochastic Activation Functions The most relevant precursor to our work is PROBACT Shridhar et al. [2019], which introduces a probabilistic activation function by sampling from a normal distribution centered at a ReLU-like mean. Unlike PROBACT, which treats neurons independently, SSA introduces a dependencies between neurons via a softmax-based categorical distribution. Other approaches like Noisy Activation Functions Gulcehre et al. [2016] inject noise specifically in saturated regimes to aid exploration. Our work differs by making the stochasticity the central mechanism of the activation itself, rather than an additive noise component.

Differentiable Categorical Selection Our methodology relies heavily on the Gumbel-Softmax trick Jang et al. [2017] and the Straight-Through Estimator Bengio et al. [2013]. These techniques have been used for discrete choice modeling and categorical reparameterization in various contexts, such as Variational Autoencoders. In the domain of activation functions, FLEX-ACT Kumar et al. [2026] uses Gumbel-Softmax to allow layers to pick from a discrete set of existing activation functions (e.g., picking between ReLU and Tanh). In contrast, we use these tools to perform sampling over the *feature space* within a single layer, rather than selecting the functional form of the layer itself.

Competitive and Sparse Representations The idea of competition between neurons has roots in lateral inhibition and "winner-take-all" mechanisms. Softmax is a standard way to implement a soft version of this competition, typically reserved for the final output. By moving this competition to intermediate layers, we relate to works on sparse coding and attention mechanisms. However, SSA is unique in its probabilistic nature and its use as a primitive replacement for standard monotonic activations. Randomized Automatic Differentiation Oktay et al. [2020] also explores stochasticity in the backward pass, providing a theoretical foundation for training networks with the high-variance gradients that often accompany sampling-based approaches.

3 Methodology

Softmax Sampling Activation (SSA) Let $\boldsymbol{x} \in \mathbb{R}^C$ be the input vector to the activation module (e.g., the output of a convolutional layer with C channels at a specific spatial location). In a standard network, an element-wise function f is applied such that $\boldsymbol{y}_i = f(\boldsymbol{x}_i)$. In contrast, SSA treats $\boldsymbol{x}$ as a vector of logits and defines a categorical distribution over the C channels.

The probability of selecting channel j is given by the softmax function:

$$P(j|\boldsymbol{x}) = \frac{\exp(\boldsymbol{x}_j/\tau)}{\sum_{k=1}^{C} \exp(\boldsymbol{x}_k/\tau)} \tag{1}$$

where τ is a temperature parameter that controls the smoothness of the distribution. As $\tau \to 0$, the distribution approaches a one-hot vector at the maximum logit, and as $\tau \to \infty$, it approaches a uniform distribution.

Differentiable Sampling with Gumbel-Softmax To train a network with categorical sampling, we use the Gumbel-Softmax reparameterization trick Jang et al. [2017]. We sample a vector of i.i.d. Gumbel noise $\boldsymbol{g}$ where $\boldsymbol{g}_i = -\log(-\log(u_i))$ and $u_i \sim \text{Uniform}(0, 1)$. The "soft" sample $\boldsymbol{s}$ is computed as:

$$\boldsymbol{s}_j = \frac{\exp((\boldsymbol{x}_j + \boldsymbol{g}_j)/\tau)}{\sum_{k=1}^{C} \exp((\boldsymbol{x}_k + \boldsymbol{g}_k)/\tau)} \tag{2}$$

The output of the SSA module is then the element-wise product of the soft sample and the original activations: $\boldsymbol{y} = \boldsymbol{s} \odot \boldsymbol{x}$. In the "hard" variant of SSA, we use the Straight-Through Estimator to perform discrete sampling in the forward pass while using the soft gradient in the backward pass.

Architectural Variants We evaluate three primary configurations of SSA:

- **Pure SSA**: Every ReLU in the network is replaced by an SSA module.
- **Mixed SSA**: Standard ReLU is used in early layers, and SSA is applied only at the final feature layer (before the global average pooling). This approach aims to preserve feature extraction while introducing stochastic competition at the decision-making stage.
- **Scaled SSA**: Since the softmax outputs sum to 1.0, the average activation value is significantly reduced compared to ReLU. In SSA-SCALED, we multiply the output by the number of channels C to maintain the original activation magnitude.

Experimental Setup We use VGG-11 Simonyan and Zisserman [2015] as our base architecture and CIFAR-10 Krizhevsky [2009] as our benchmark dataset. All models are trained for 10 epochs using SGD with a learning rate of 0.01, momentum of 0.9, and a cosine annealing scheduler. We use 4x NVIDIA RTX A6000 GPUs for all experiments. The temperature τ is fixed at 1.0 for all SSA modules.

4 Results

Our experimental results are summarized in table 1. We compare the baseline VGG-11 with standard ReLU activations against several variants of SSA.

Table 1: Main results on CIFAR-10 using VGG-11. We report Top-1 Test Accuracy and Expected Calibration Error (ECE). Best results in **bold**.

Method	Test Accuracy (%)	ECE	Status
ReLU Baseline	**86.46**	0.0236	Success
SSA (All Layers, Soft)	10.21	0.0053	Failed
SSA (All Layers, Hard)	14.23	0.0041	Failed
MIXED-SSA (Last Layer, Soft)	**83.09**	0.0323	Success
MIXED-SSA (Last Layer, Hard)	10.38	0.0713	Failed

Failure of Pure SSA Stacks A striking finding is the complete failure of networks where every activation is replaced by SSA. Both "soft" and "hard" variants failed to exceed 15% accuracy, which is only marginally better than random guessing. As shown in figure 1, these models show virtually no progress during the 10 epochs of training. We hypothesize that this is due to the "vanishing signal" problem: because the softmax outputs at each layer sum to 1.0, the average activation value is scaled down by a factor of $1/C$ at each layer. In a deep network like VGG-11, this attenuation becomes extreme, effectively zeroing out the signal and the gradients.

Effectiveness of Hybrid Architectures In contrast, applying SSA only at the final feature layer (MIXED-SSA) proved to be much more stable. The "soft" variant achieved 83.09% accuracy, which is a respectable performance given the short training duration. This demonstrates that categorical sampling is a viable primitive when used as a specialized bottleneck rather than a general-purpose activation function. However, the "hard" variant of MIXED-SSA still failed to train, indicating that the high variance of discrete sampling gradients is particularly disruptive in competitive distributions.

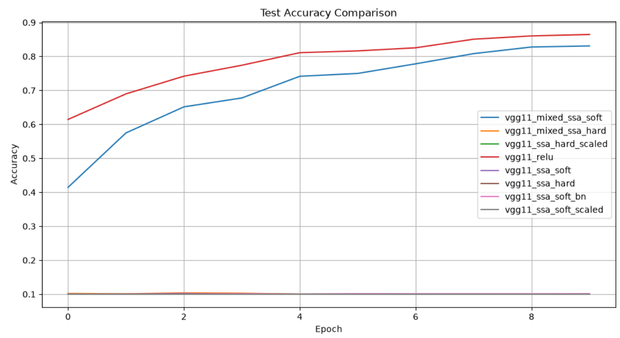

Figure 1: Test accuracy over 10 epochs of training on CIFAR-10. While the ReLU baseline and MIXED-SSA (Soft) converge, all variants using SSA at every layer or using ”hard” sampling fail to show significant progress.

Uncertainty Calibration We also measured the Expected Calibration Error (ECE) to see if the stochasticity of SSA improves uncertainty quantification. Interestingly, MIXED-SSA (Soft) showed a slightly higher ECE (0.0323) than the ReLU baseline (0.0236). This suggests that while SSA introduces stochasticity, it does not automatically lead to better-calibrated probabilities, at least without further tuning of the temperature parameter.

5 Discussion

Interpretability and Competitive Bottlenecks The core mechanism of SSA is categorical competition. Unlike ReLU, which allows multiple independent features to be active simultaneously, SSA forces neurons to ”compete” for a fixed amount of activation probability (summing to 1.0). Our results suggest that this ”winner-take-all” behavior acts as an information bottleneck. While this can serve as a powerful regularizer by forcing the network to focus on the most salient features, it also restricts the model’s total representational capacity. In tasks like image classification, where multiple hierarchical features often co-exist, the strict competition of SSA might be overly restrictive.

Signal Propagation in Stochastic Networks The ”vanishing signal” problem we observed in deep SSA stacks highlights a fundamental challenge in designing stochastic primitives. Monotonic activations like ReLU are designed to preserve signal magnitude and gradient flow across many layers. In contrast, softmax-based activations are inherently contractive. For SSA to be used more broadly, specialized initialization schemes or residual connections (e.g., $y = \text{ReLU}(x) + \text{SSA}(x)$) may be necessary to ensure that the signal can penetrate deep architectures without decaying.

Soft vs. Hard Sampling The dramatic performance gap between ”soft” relaxation and ”hard” sampling in our experiments underscores the difficulty of training with discrete categorical choices. Even with the Straight-Through Estimator, the variance of the gradients in a high-dimensional categorical space seems to be too high for standard optimizers to handle during the early stages of training. This suggests that ”annealing” strategies—where the temperature τ starts high (soft) and gradually decreases (hard)—might be essential for unlocking the potential of discrete sampling activations.

Limitations Our study was limited by a relatively short training duration (10 epochs) and a focus on a single architecture (VGG-11). It is possible that with longer training or more modern architectures like ResNet, pure SSA stacks might show better convergence. Additionally, we did not perform a hyperparameter search over the temperature τ, which likely plays a critical role in balancing competition and gradient flow.

6 Conclusion

In this paper, we introduced SSA (SSA), a novel activation module that brings the categorical sampling of the final softmax layer into the intermediate representations of a neural network. We demonstrated that while deep stacks of competitive activations suffer from severe signal attenuation, a selective hybrid approach can achieve performance comparable to standard deterministic baselines. Our findings highlight the importance of continuous relaxation (Gumbel-Softmax) and the trade-offs between independent feature representation and categorical competition.

Future work will focus on temperature annealing schedules to bridge the gap between soft and hard sampling, as well as exploring SSA as an additive component in residual networks. We also believe that SSA may find applications in sparse attention mechanisms and interpretable feature selection, where a probabilistic bottleneck can provide valuable insights into the model's internal decision-making process.

References

Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. In *arXiv preprint arXiv:1308.3432*, 2013.

Caglar Gulcehre, Marcin Moczulski, Misha Denil, and Yoshua Bengio. Noisy activation functions. In *International Conference on Machine Learning*, 2016.

Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with gumbel-softmax. In *International Conference on Learning Representations*, 2017.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Citeseer, 2009.

Ramnath Kumar, Kyle Ritscher, and et al. Flex-act: Why learn when you can pick? In *arXiv preprint arXiv:2601.06441*, 2026.

Deniz Oktay, Nick McGreivy, and et al. Randomized automatic differentiation. In *arXiv preprint arXiv:2007.10412*, 2020.

Kumar Shridhar, Joonho Lee, Hideaki Hayashi, and et al. Probact: A probabilistic activation function for deep neural networks. In *arXiv preprint arXiv:1905.10761*, 2019.

Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations*, 2015.

Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. In *Journal of Machine Learning Research*, 2014.